Junhao Huang

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Work

- **Singapore Management University** — Supervisor: Dr. Haiyang Xue
  Research Scientist — *Jan. 2026-Now*
- **City University of Hong Kong** — Supervisor: Prof. Ray C.C. Cheung
  Postdoc — *Oct. 2025-Dec. 2025*

Education

- **Beijing Normal-Hong Kong Baptist University** — Supervisor: Dr. Donglong Chen
  PhD Degree of Computer Science (Hong Kong Baptist University) — *Sep. 2021-Jul. 2025*
- **Nanjing University of Aeronautics and Astronautics** — Supervisor: Prof. Zhe Liu
  Master Degree of Cyberspace Security — *Sep. 2018-Jun. 2021*
- **Nanjing University of Aeronautics and Astronautics** — GPA: 3.7
  Bachelor Degree of Computer Science and Technology — *Sep. 2014-Jun. 2018*

Research/Visiting Activities

- **Visiting Scholar, Electrical Engineering** — Hong Kong, China
  *City University of Hong Kong (**Prof. Ray C. C. Cheung**)* — *Jul. 2023-Dec. 2023*
- **Visiting Scholar, Cyberspace Security** — Whuhan, China
  *Wuhan University (**Prof. Debiao He**)* — *Sep. 2019-Jan. 2020*

Research Interests

- Modular Arithmetic, Post-quantum Cryptography, Fully Homomorphic Encryption, Embedded Security.

Scientific Services

I have been actively participating as a Program Committee/reviewer for the following conferences/journals:

- *IACR Conference on Cryptographic Hardware and Embedded Systems (CHES 2027)*
- *CCF International Conference on Service Science 2026 (CCF ICSS 2026)*

- *ACM Transactions on Architecture and Code Optimization (TACO).*
- *Journal of Cryptographic Engineering (JCEN).*
- *The Journal of Supercomputing*.
- *Artifact Evaluation Committee of IACR CHES 2024∼ 2026*

Representative Publications

1. **Junhao Huang**, Yijian Liu, Yu Zhang, Xianhui Lu, Donglong Chen, Haiyang Xue*, DAWN′: Improved DAWN and Accelerated Implementation on ARM Cortex-M4, *IACR Transactions on Cryptographic Hardware and Embedded Systems, 2027.*
2. **Junhao Huang**, Yiteng Sun, Jipeng Zhang, Weijia Wang, Donglong Chen*, Haiyang Xue, Guomin Yang, Efficient High-order Masking Raccoon on ARM Cortex-M4, *IACR Transactions on Cryptographic Hardware and Embedded Systems, 2026.*
3. Haoyang Chen, **Junhao Huang**, Shutong Jin, Ray C. C. Cheung, Donglong Chen, Wangchen Dai*, High-Throughput GPU Design and Implementation of MAYO with Matrix Computation Reordering, *IACR Transactions on Cryptographic Hardware and Embedded Systems, 2026.*
4. Haosong Zhao, **Junhao Huang**, Donglong Chen, Renjing Xu, Hongyuan Liu*, 1MMED1ATE: Private Tree Ensemble Inference via One Homomorphic Matrix-Vector Multiplication, *ACM SIGMETRICS 2027.*
5. **Junhao Huang**, Haosong Zhao, Jipeng Zhang, Wangchen Dai, Lu Zhou, Çetin Kaya Koç, Ray C.C. Cheung, Donglong Chen*, Yet another Improvement of Plantard Arithmetic for Faster Kyber on Low-end 32-bit IoT Devices, *IEEE Transactions on Information Forensics & Security, 2024.*
6. **Junhao Huang**, Alexandre Adomnicăi, Jipeng Zhang, Wangchen Dai, Yao Liu, Ray C. C. Cheung, Çetin Kaya Koç, Donglong Chen*, Revisiting Keccak and Dilithium Implementations on ARMv7-M, *IACR Transactions on Cryptographic Hardware and Embedded Systems, 2024.*
7. **Junhao Huang**, Jipeng Zhang, Haosong Zhao, Zhe Liu, Ray C. C. Cheung, Çetin Kaya Koç, Donglong Chen*, Improved Plantard Arithmetic for Lattice-based Cryptography, *IACR Transactions on Cryptographic Hardware and Embedded Systems, 2022.*
8. **Junhao Huang**, Zhe Liu*, Zhi Hu, and Johann Großschädl, Parallel Implementation of SM2 Elliptic Curve on Intel Processor with AVX2, *Australasian Conference on Information Security and Privacy - ACISP 2020.*
9. Zewen Ye, **Junhao Huang**, Tianshun Huang, Yudan Bai, Jinze Li, Hao Zhang, Guangyan Li, Donglong Chen, Ray CC Cheung, Kejie Huang, PQNTRU: Acceleration of NTRU-based Schemes via Customized Post-Quantum Processor. *IEEE Transactions on Computers, 2025*
10. Haosong Zhao, **Junhao Huang**, Zihang Chen, Kunxiong Zhu, Donglong Chen, Zhuoran Ji, Hongyuan Liu, VESTA: A Secure and Efficient FHE-based Three-Party Vectorized Evaluation System for Tree Aggregation Models. *ACM SIGMETRICS, 2025*

11. Jipeng Zhang, Yuxing Yan, **Junhao Huang**, Çetin Kaya Koç*, Optimized Software Implementation of Keccak, Kyber, and Dilithium on RV{32,64}IM{B}{V}, *IACR Transactions on Cryptographic Hardware and Embedded Systems, 2025.*
12. Jipeng Zhang, **Junhao Huang**, Lirui Zhao, Donglong Chen, Çetin Kaya Koç*, ENG25519: Faster TLS 1.3 handshake using optimized X25519 and Ed25519, *Usenix Security, 2024.*
13. Jipeng Zhang, **Junhao Huang**, Zhe Liu*, Sujoy Sinha Roy, Time-memory Trade-offs for Saber on Memory-constrained RISC-V, *IEEE Transactions on Computers, 2022.*
14. Xinyi Ji, Jiankuo Dong, **Junhao Huang**, Zhijian Yuan, Wangchen Dai, Fu Xiao, Jingqiang Lin, ECO-CRYSTALS: Efficient Cryptography CRYSTALS on Standard RISC-V ISA, *IEEE Transactions on Computers, 2024.*

ICCS NGCC Proposals

I have participated in the efficient implementations of the following NGCC proposals:

- ZEN: Zero Divisor Encoding on NTRU Encryption, Xianhui Lu, Yijian Liu, Dongshu Cai, Dingding Jia, Lei Bi, Jingnan He, Yu Zhang, Zhe Liu, **Junhao Huang**, Lu Zhou, Liming Fang, Tianhao Liu, Sihan He, Meng Xu, Lutan Zhao, Xueqian Bao, *Institute of Commercial Cryptography Standards*, 2026.
- DKEM (Ding Key Encapsulation), Rui Liu, Chen Kong, Wenbin Zhang, Neng Zeng, Han Xia, Manoj Gyawali, Samuel José García, Alba Hernández Costoya, Haiyang Xue, **Junhao Huang**, Boru Gong, Yi Jin, *Institute of Commercial Cryptography Standards*, 2026.
- DKEX (Ding Key Exchange), Rui Liu, Chen Kong, Wenbin Zhang, Neng Zeng, Han Xia, Manoj Gyawali, Samuel José García, Alba Hernández Costoya, Haiyang Xue, **Junhao Huang**, Boru Gong, Yi Jin, *Institute of Commercial Cryptography Standards*, 2026.
- ADKEX (Authenticated Ding Key Exchange), Rui Liu, Chen Kong, Wenbin Zhang, Neng Zeng, Han Xia, Manoj Gyawali, Samuel José García, Alba Hernández Costoya, Haiyang Xue, **Junhao Huang**, Boru Gong, Yi Jin, *Institute of Commercial Cryptography Standards*, 2026.

Academic Referee

1. **Dr. Donglong Chen**: Associate Professor of Beijing Normal-Hong Kong Baptist University, donglongchen@uic.edu.cn
2. **Prof. Ray C.C. Cheung**: Associate Provost, Professor of City University of Hong Kong, r.cheung@cityu.edu.hk
3. **Prof. Çetin Kaya Koç**: IEEE Life Fellow, Co-founder of CHES, University of California Santa Barbara, cetinkoc@ucsb.edu